

Timber workshops — ground-screw foundation & floor plan

Two timber workshops on ground screws. Estimated ground-screw count & positions, and 5"×2" joist / bearer quantities. Prepared 30 Jul 2026.

Footprints

	Workshop 1	Workshop 2
Footprint	6.0 m × 4.0 m (rectangle)	~4.0 m × ~7.2 m, bottom-left corner clipped
Internal area	22.06 m²	24.46 m²

Design assumptions

- **Joists:** 5"×2" (47×125 mm), ideally C24 graded, at **400 mm centres**.
- **Max joist span 2.0 m** between supports — 5×2 at 400 c/c is at its limit at 2 m. For heavy machinery, tighten support lines to 1.5 m or step up to 6×2.
- Joists sit on **bearer beams** (5×2), which sit on the ground-screw heads — this is the standard method and it minimises the number of (costly) screws.
- Ground screws with **adjustable U-head / plate tops**, at ≤1.5 m centres along each bearer.
- Deck on top (not in this take-off): 18–22 mm OSB3 T&G.

Headline quantities

- **Ground screws:** ~33 total — 15 for Workshop 1, ~18 for Workshop 2 (order ~36 with spares).
- **5×2 timber:** ~215 linear metres of framing — call it ~240 m allowing for waste & noggins.

Workshop 1 — 6.0 × 4.0 m

- **Ground screws:** 15 — grid at X = 0, 1.5, 3.0, 4.5, 6.0 m × Y = 0, 2.0, 4.0 m.
- **Bearers:** 3 runs × 6.0 m (along the length) = **18 m**.
- **Joists:** 16 × 4.0 m (spanning the 4 m width, 400 c/c) = **64 m**.
- **Noggins** over the 3 bearer lines: **18 m**. Room total 5×2 ≈ **100 m**.

Workshop 2 — ~4.0 × 7.2 m (clipped corner)

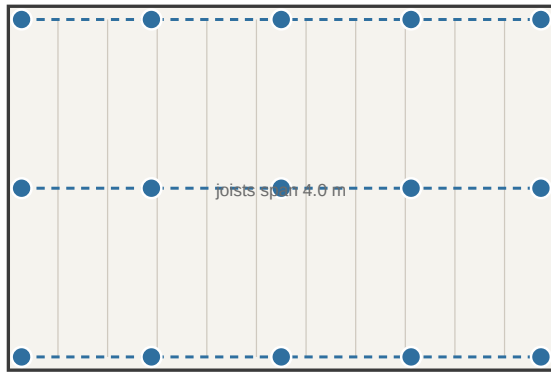
- **Ground screws:** ~18 — 3 bearer lines at X = 0, 2.0, 4.0 m; screws at Y = 0, 1.5, 3.0, 4.5, 6.0, 7.2 m. Left line cut short by the angled wall, plus 2 extra at the door threshold.
- **Bearers:** ≈ **21 m** (left run shortened by the clip).
- **Joists:** ~19 × 4.0 m ≈ **72–76 m** (a few short at the clip).
- **Noggins:** ~21 m. Room total 5×2 ≈ **114 m**.

Shopping list

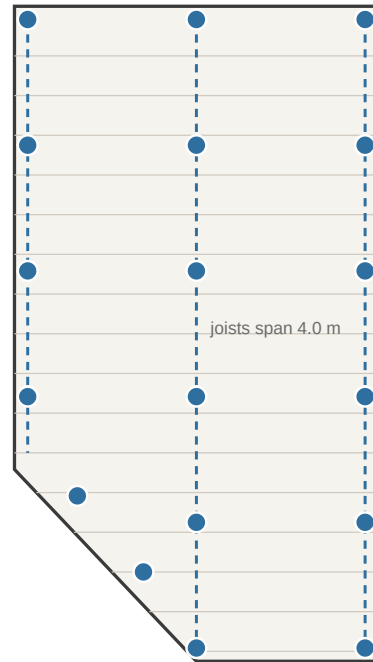
Item	Quantity	Notes
5×2 joists	35 × 4.2 m lengths	WS1 16 + WS2 19, each ~4.0 m
5×2 bearers + noggins	~18 × 4.8 m lengths	cut to 6.0 / 7.2 m runs + blocking
5×2 total	≈ 53 lengths	~240 lin. m incl. waste
Ground screws	36 (33 + 3 spares)	adjustable U-head / plate top

Ground-screw & joist layout

Workshop 1 - 6.0 x 4.0 m - 15 ground screws



Workshop 2 - ~4.0 x 7.2 m - 18 ground screws



clipped corner = angled door wall

Before you order

- **Screw length depends on the ground, not the building.** Soft / made-up ground or clay needs longer screws; firm ground shorter. Match the supplier load ratings to the ≤ 1.5 m spacing used here.
- **Confirm Workshop 2's exact shape.** The clip is estimated as a ~2 m triangle to hit the 24.46 m² internal area — give the true corner points and the WS2 counts can be tightened.
- **Confirm the load.** 5x2 at 400 c/c over 2.0 m is at its limit. For a car, big lathe, or similar, drop bearer lines to 1.5 m (adds ~1 screw per line) or use 6x2.

Indicative take-off for planning only — not a structural design. Verify spans, screw model, length and spacing against the ground-screw supplier's load tables and your actual ground conditions before building.